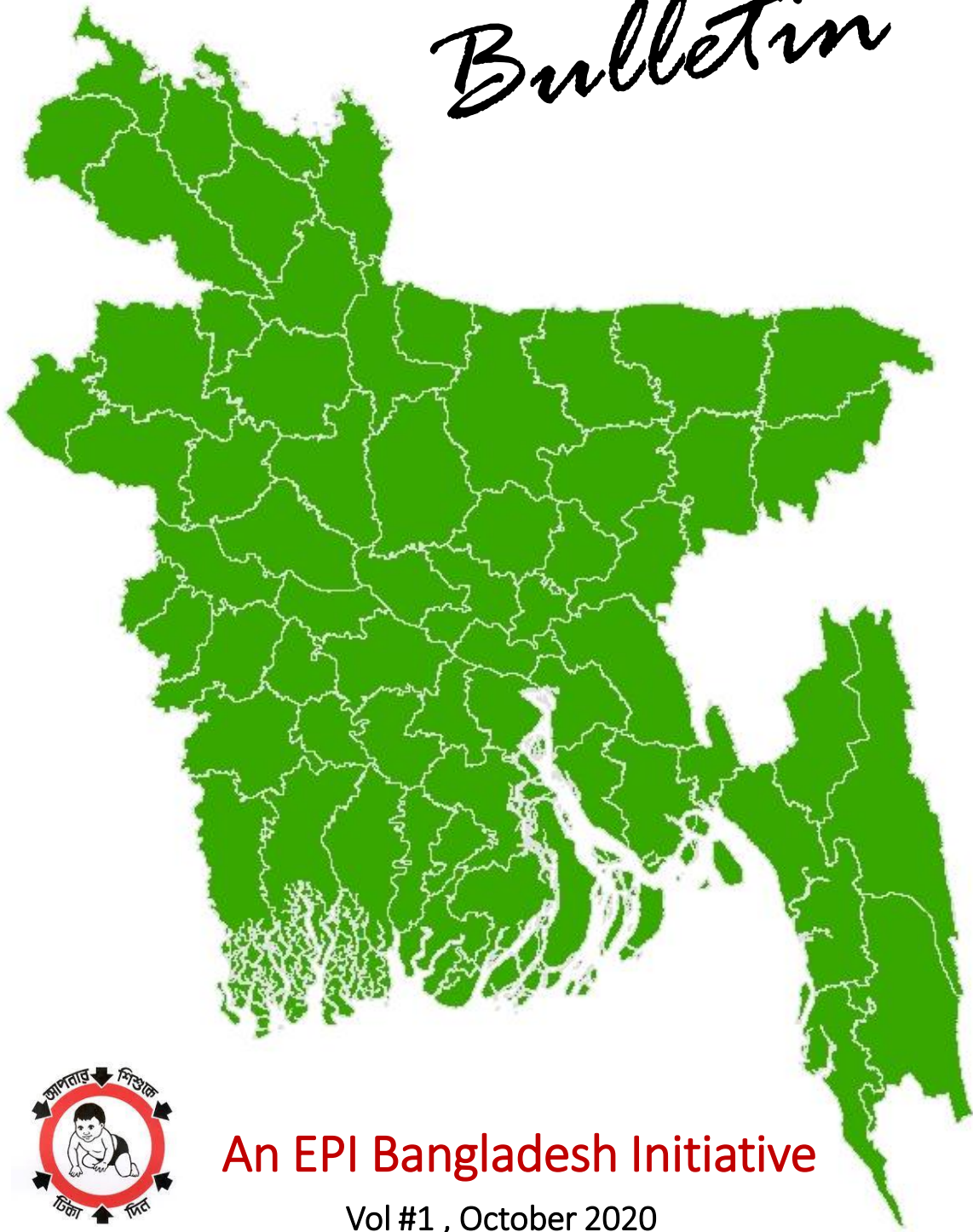




MEASLES-RUBELLA

Bulletin



An EPI Bangladesh Initiative

Vol #1 , October 2020

Initiative supported by WHO Bangladesh

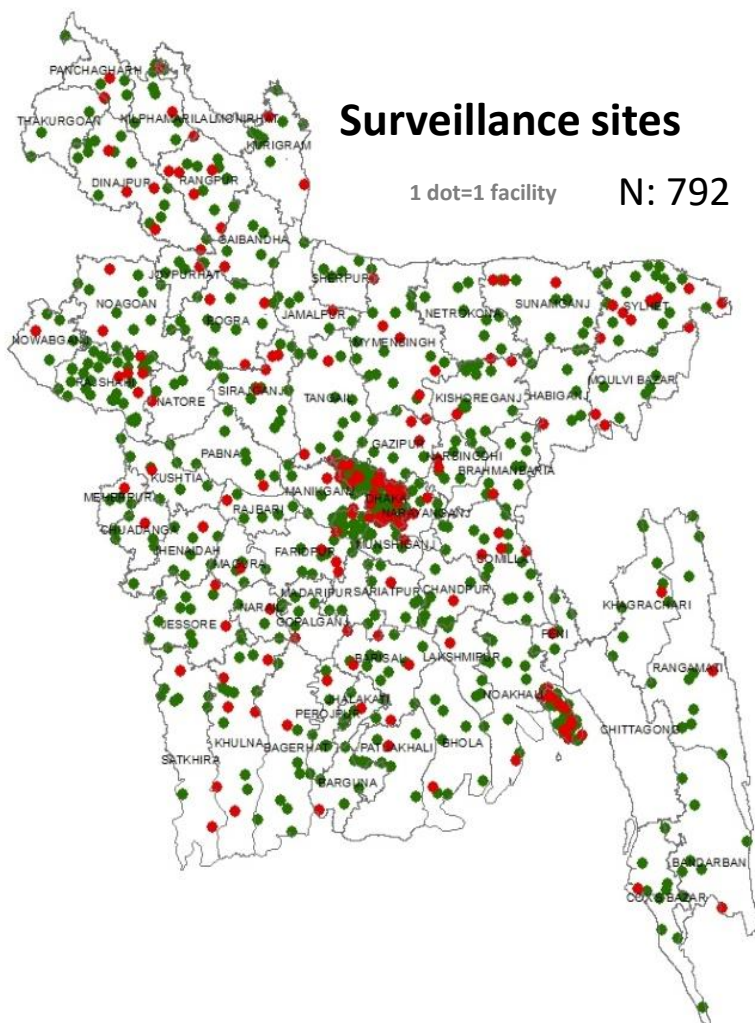


MEASLES-RUBELLA

Bulletin

Key Highlights

- 1979 : Bangladesh introduced first dose of measles vaccine
- 2003 : Laboratory supported measles surveillance initiated
- Conducted nationwide MCV campaign in 2005/2006, 2010 and 2014
- 2012 : MR second dose introduced
- 2015 : National Verification Committee (NVC) for measles, rubella & CRS elimination formed
- 2017 : Strategic Plan for the Elimination of Measles, Rubella and CRS in Bangladesh
- 2018 : Rubella & CRS controlled certification
- The WHO SEAR Region has adopted the goal of “measles and rubella elimination by 2023”
<https://apps.who.int/iris/bitstream/handle/10665/327923/sea-rc72-r3-eng.pdf?sequence=1&isAllowed=y>
- 2020 : Significant impact of COVID-19 on measles and rubella activities
- Measles-Rubella SIAs planned in quarter 1, 2020 postponed. EPI in consultations with partners to explore possibility of re scheduling the SIA based on COVID-19 scenario



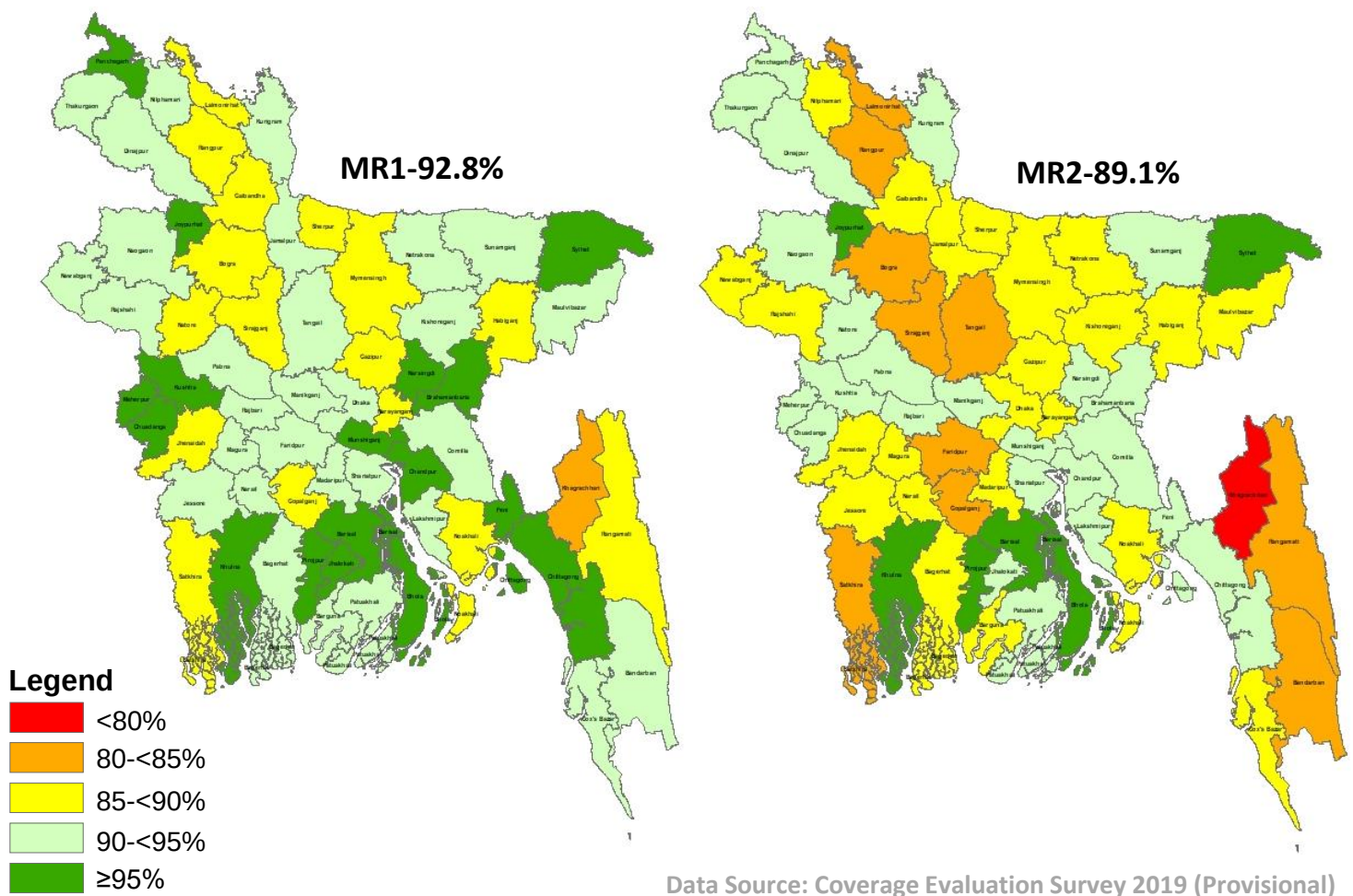
Bangladesh : Surveillance sites

Total Active surveillance sites 167

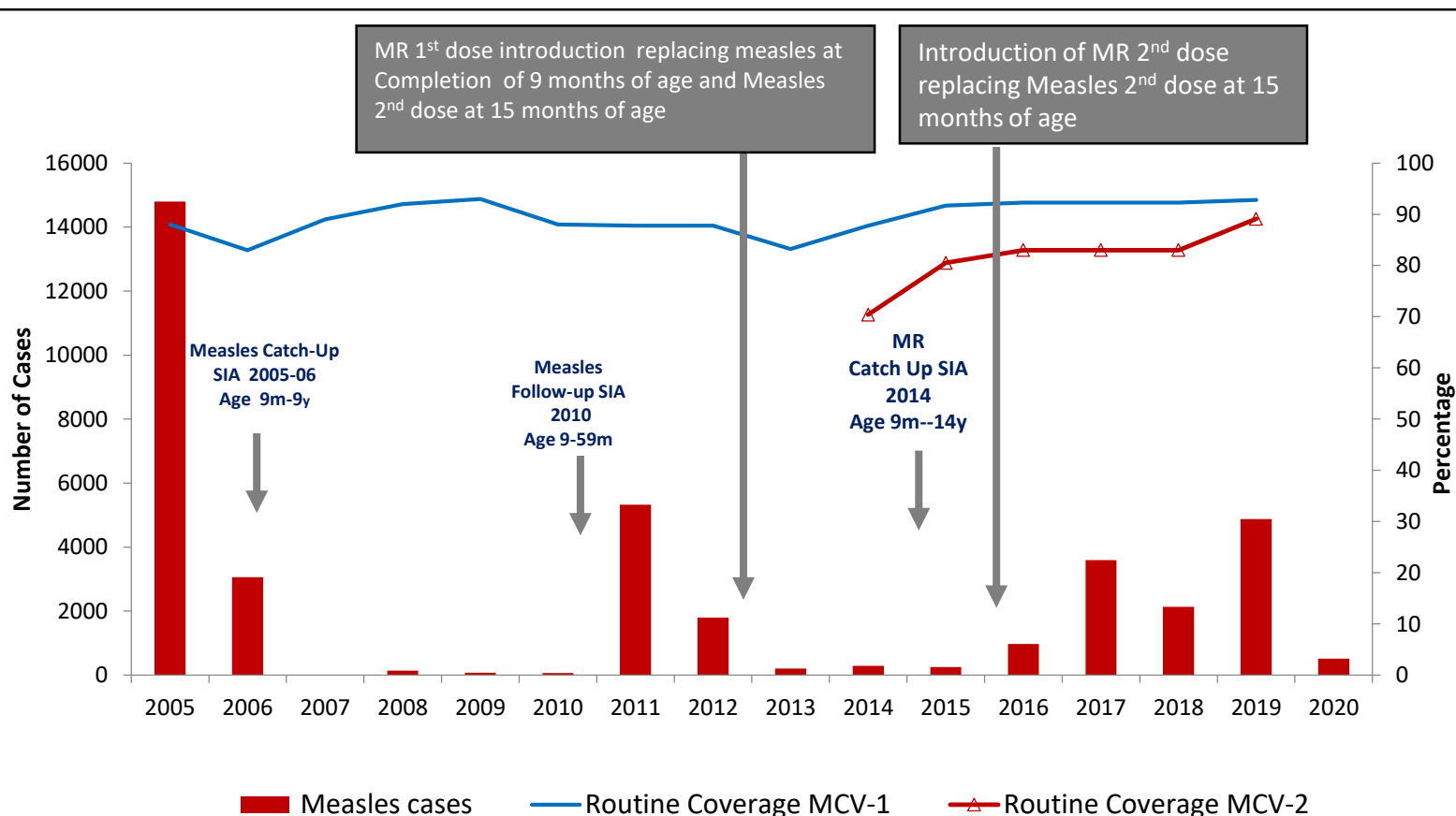
Total Passive surveillance sites* 792

* Also Includes active surveillance sites.

CES 2019 : Valid MR1 & MR2 vaccination coverage by age of 23 months

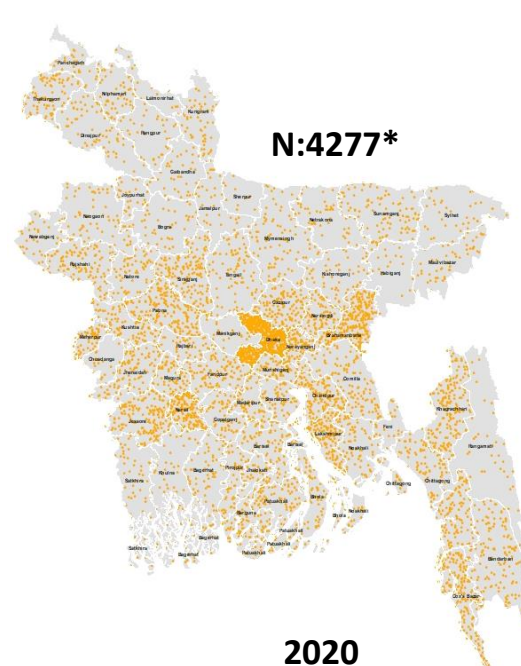
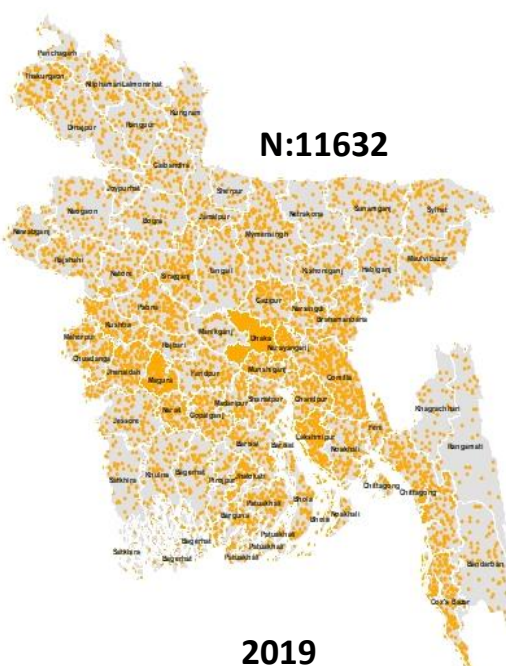
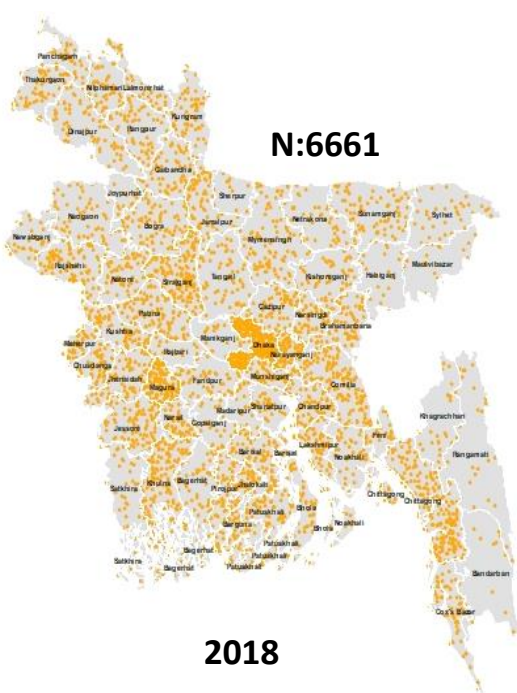


Impact of routine and supplementary vaccination on reporting of measles cases, 2005-2020*



Distribution of Suspected Measles Cases 2018, 2019 & 2020 up to September

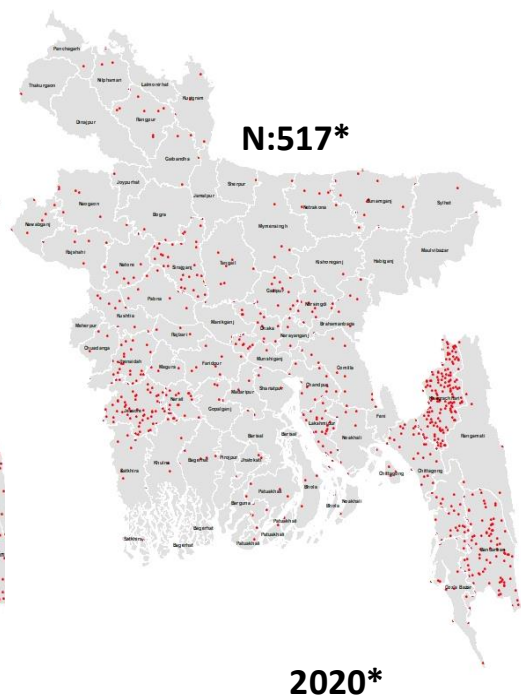
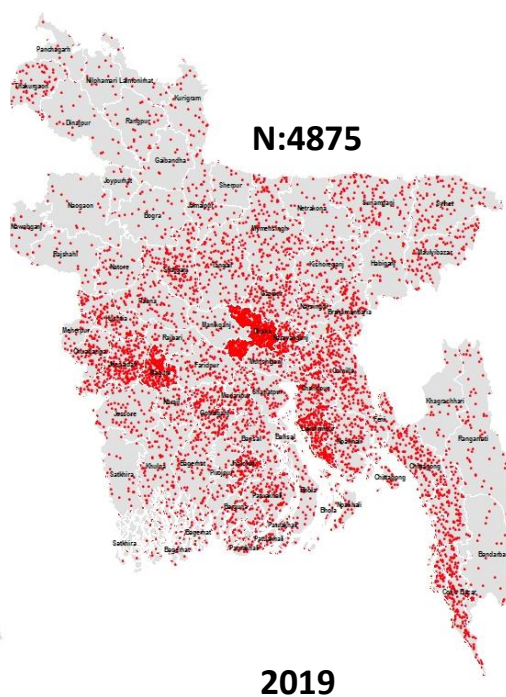
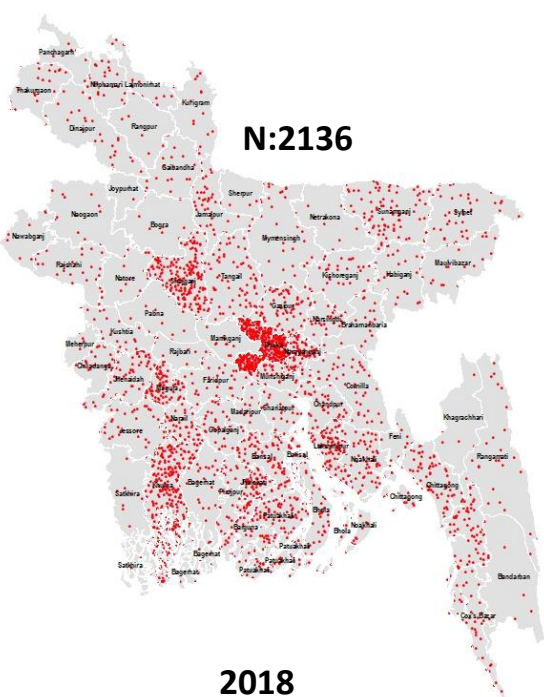
1 dot= 1 case



Dots are randomly placed within district

Distribution of Confirmed Measles Cases 2018, 2019 & 2020 up to September

1 dot= 1 case

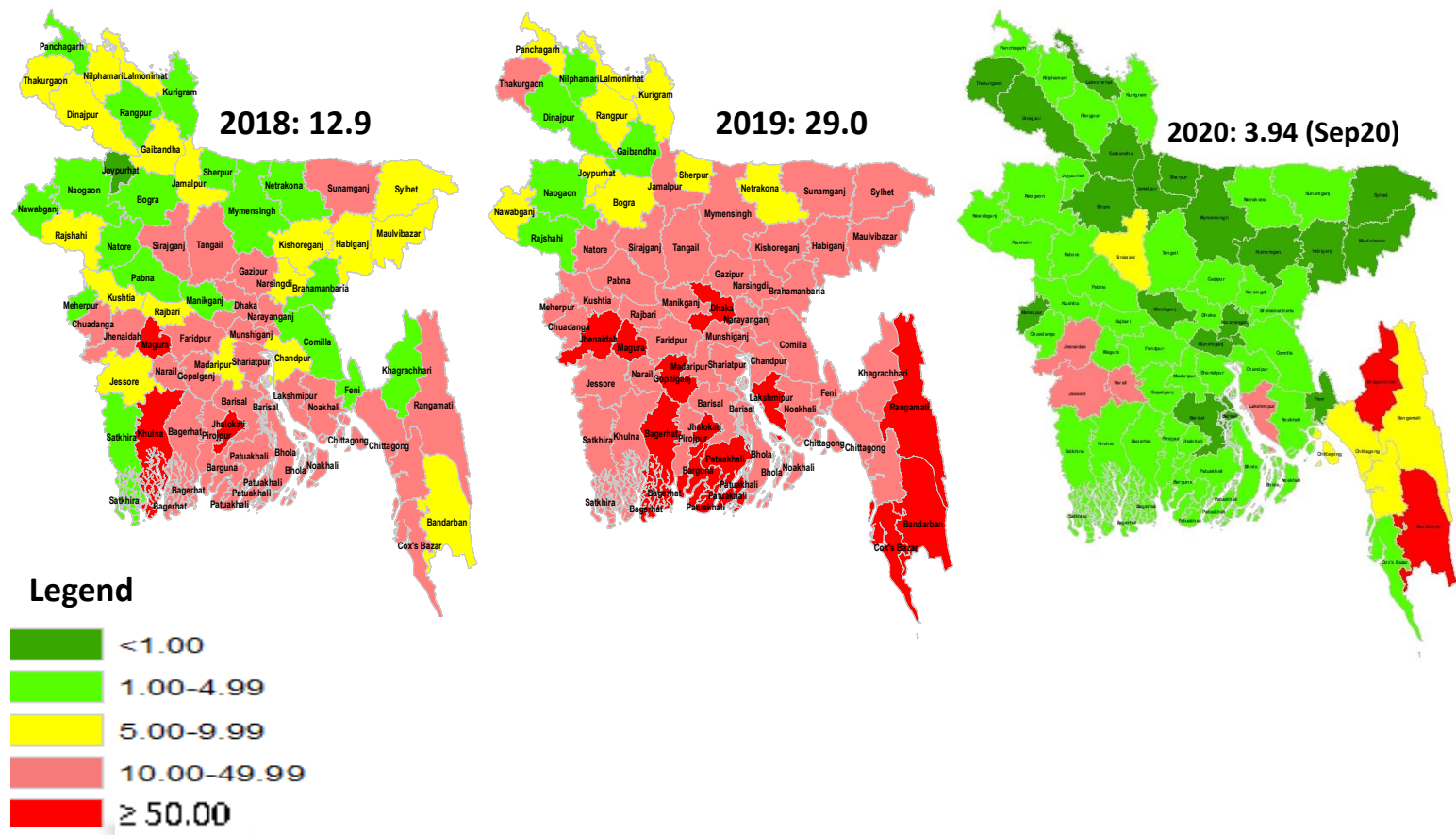


Confirmed Measles=Lab confirmed+Epi-linked cases

Dots are randomly placed within district

* Progress affected due to COVID-19

Measles Incidence per million population 2018-2020

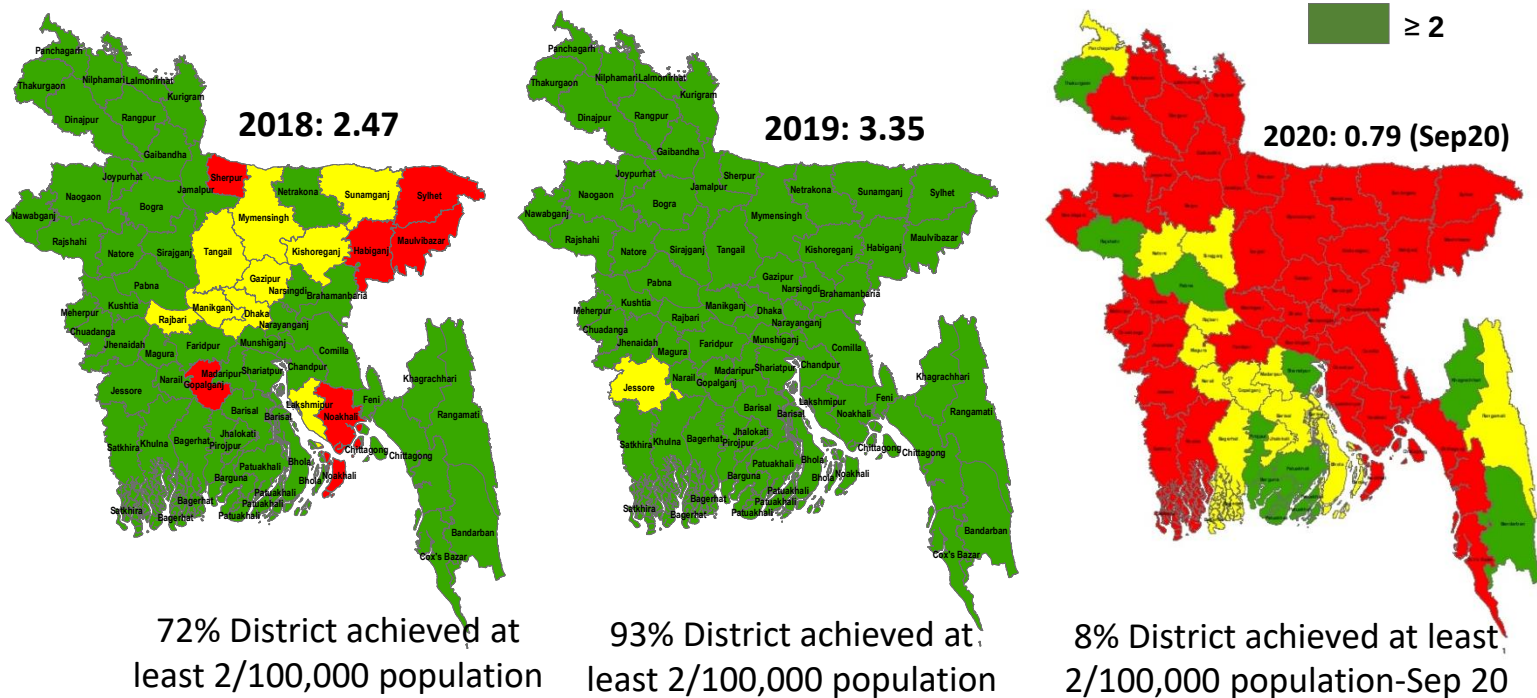


Non-measles non-rubella discarded rate per 100,000 population

National target: At least 2/100,000 population in > 80% district

Legend

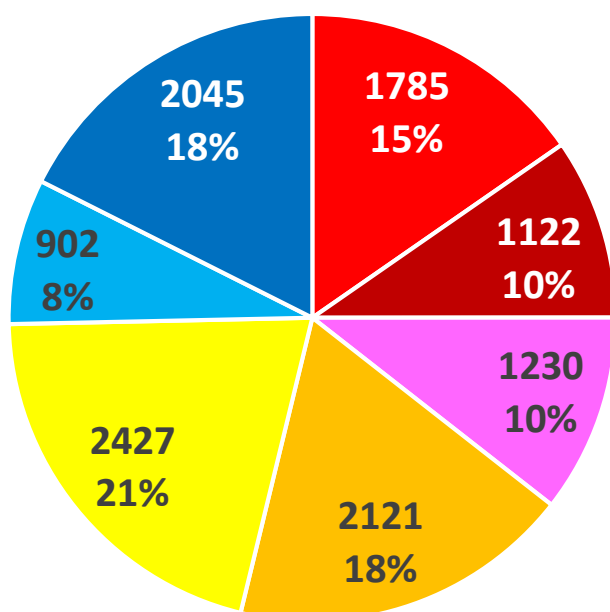
- <1
- 1-2
- ≥ 2



Age Distribution of Suspected Measles Cases

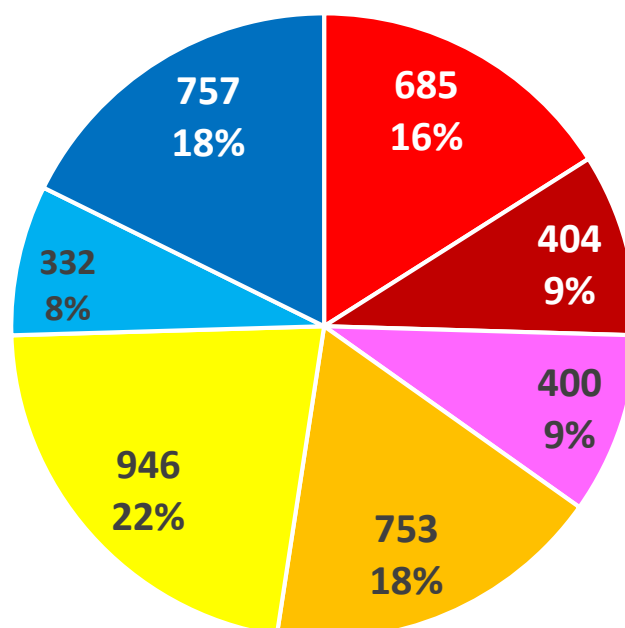
2019

N:11632



2020 (Up to September)

N:4277



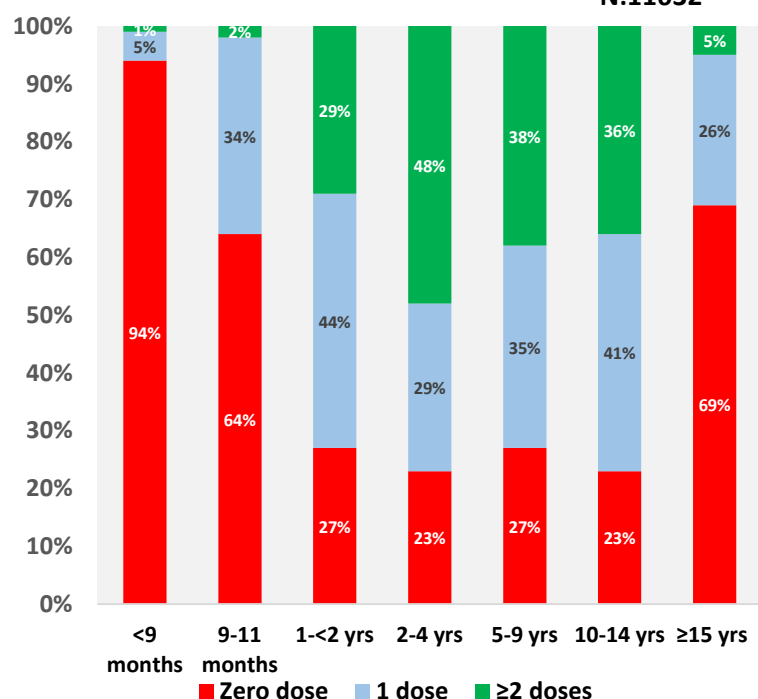
~ One third children affected are under 2 years of age

■ <9 months ■ 9-11 months ■ 1-2 yrs ■ 2-4 yrs ■ 5-9 yrs ■ 10-14 yrs ■ ≥15 yrs

Vaccination Status in Suspected Measles Cases

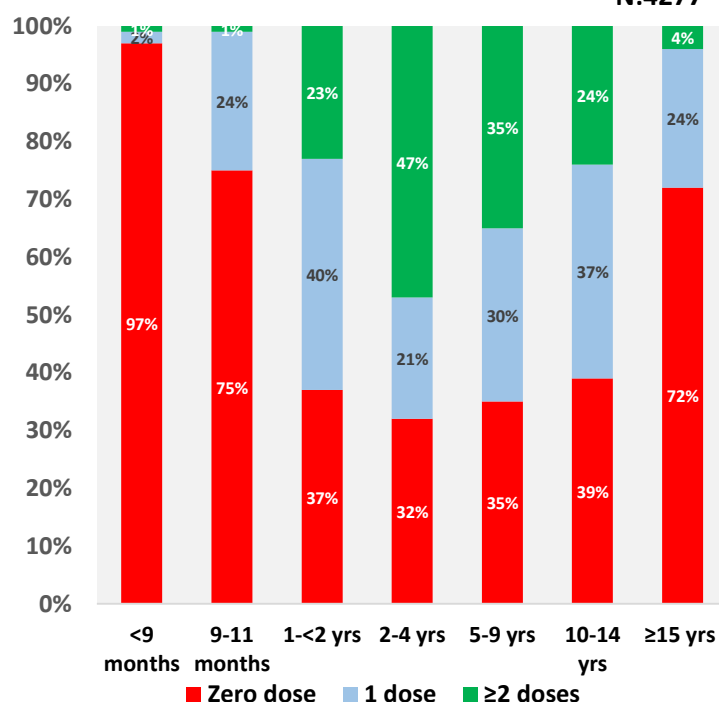
2019 (Jan-Dec)

N:11632



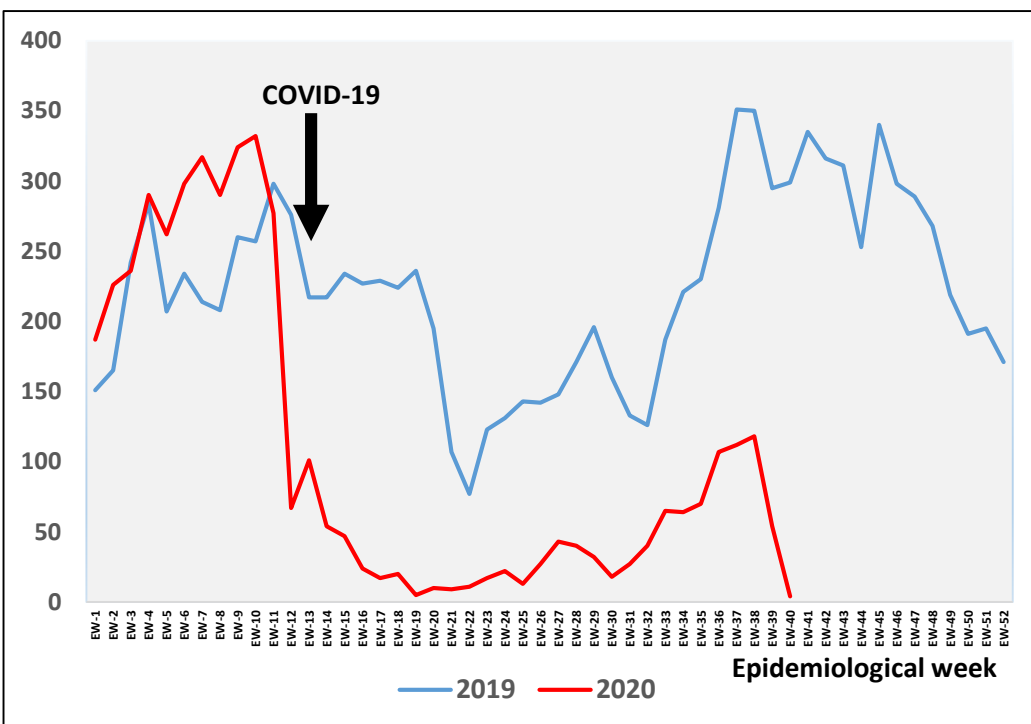
2020 (Jan-Sep)

N:4277



~ 60-65 cases under 2 years of age did not receive desired measles doses

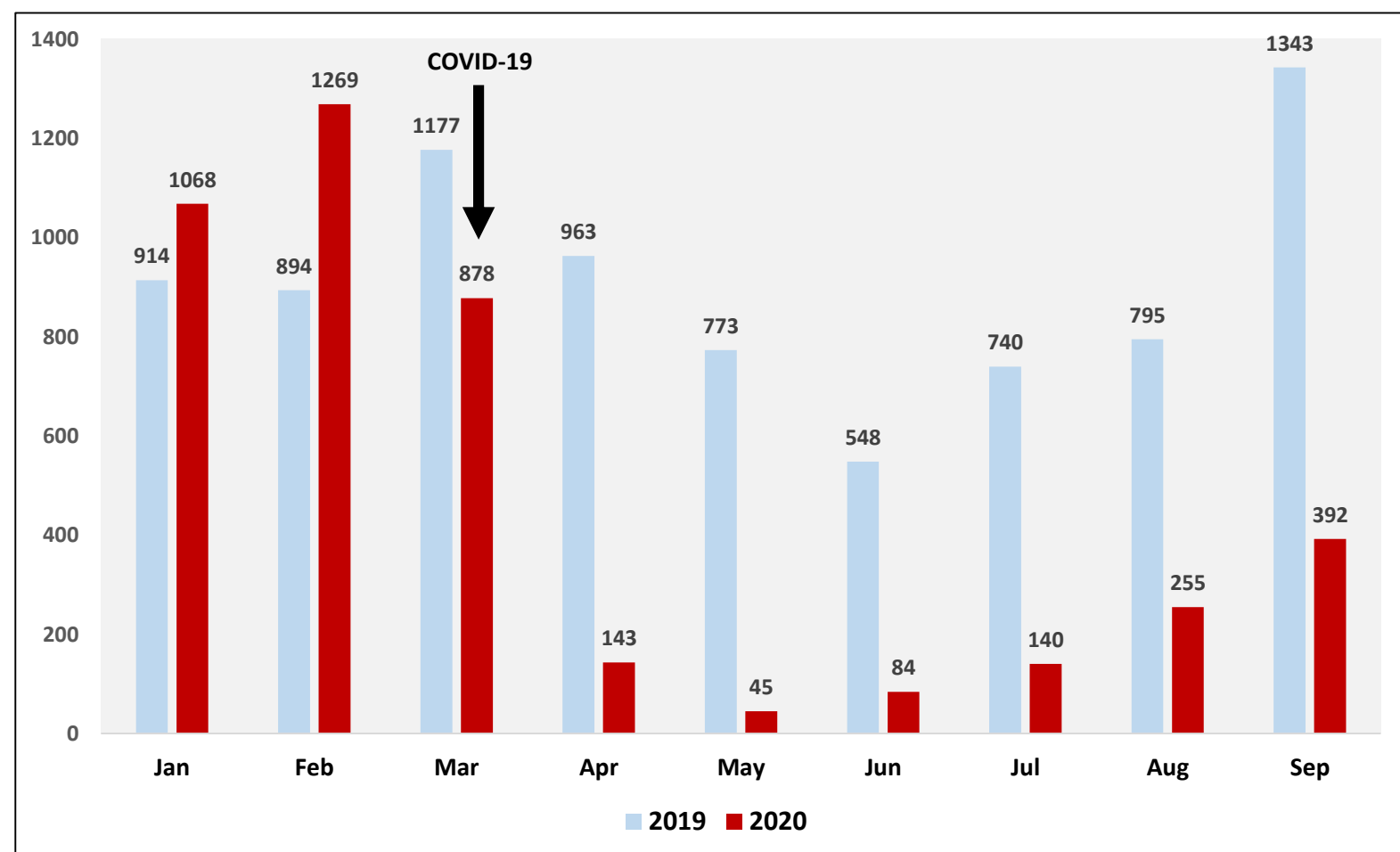
Effect of Covid-19 on suspected measles cases reported



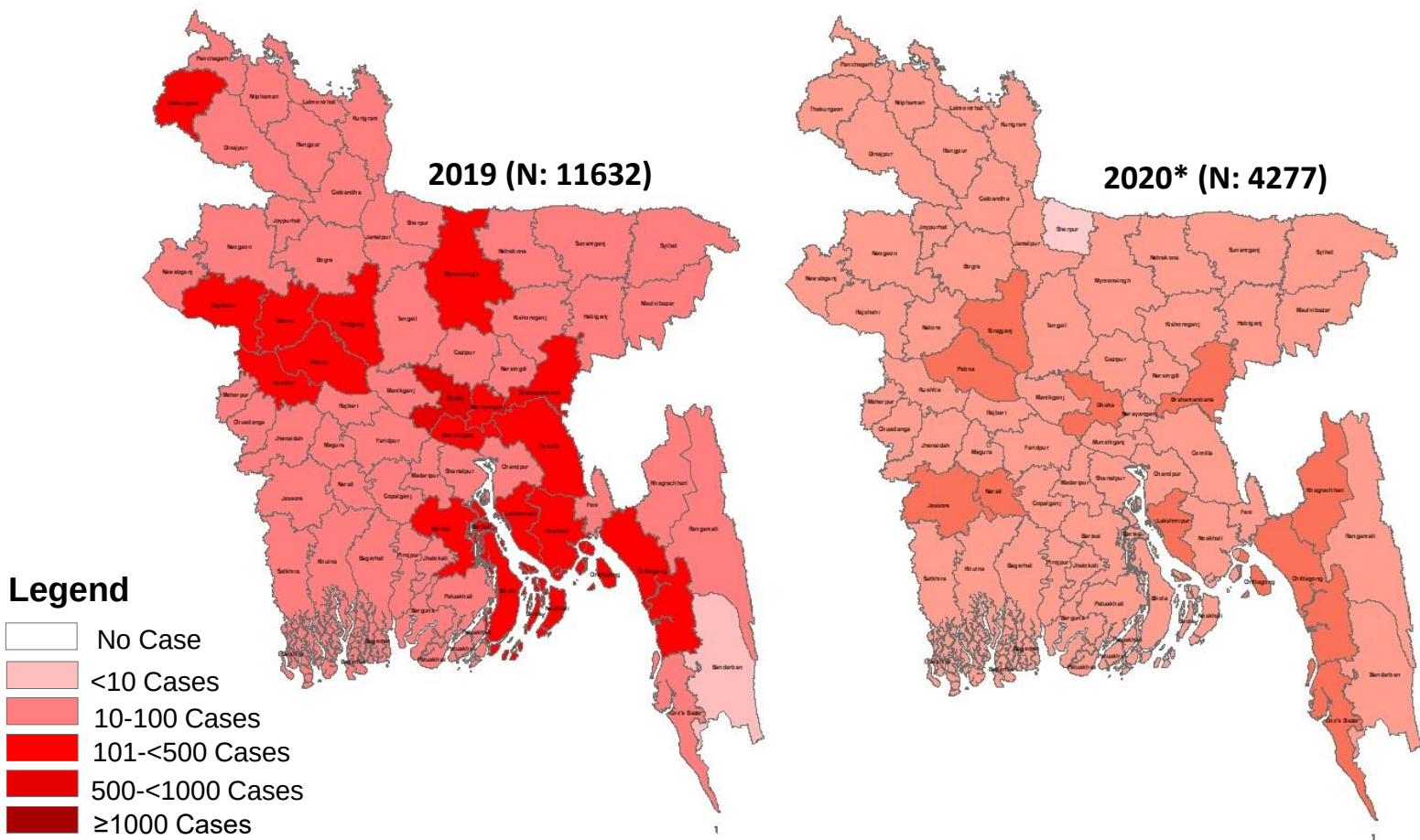
Division	2019	2020*
Barisal	950	352
Chittagong	2667	1089
Dhaka	2954	1260
Khulna	1768	608
Rajshahi	1199	505
Rangpur	922	290
Sylhet	607	173
Total	11632	4277

* Progress affected due to COVID-19

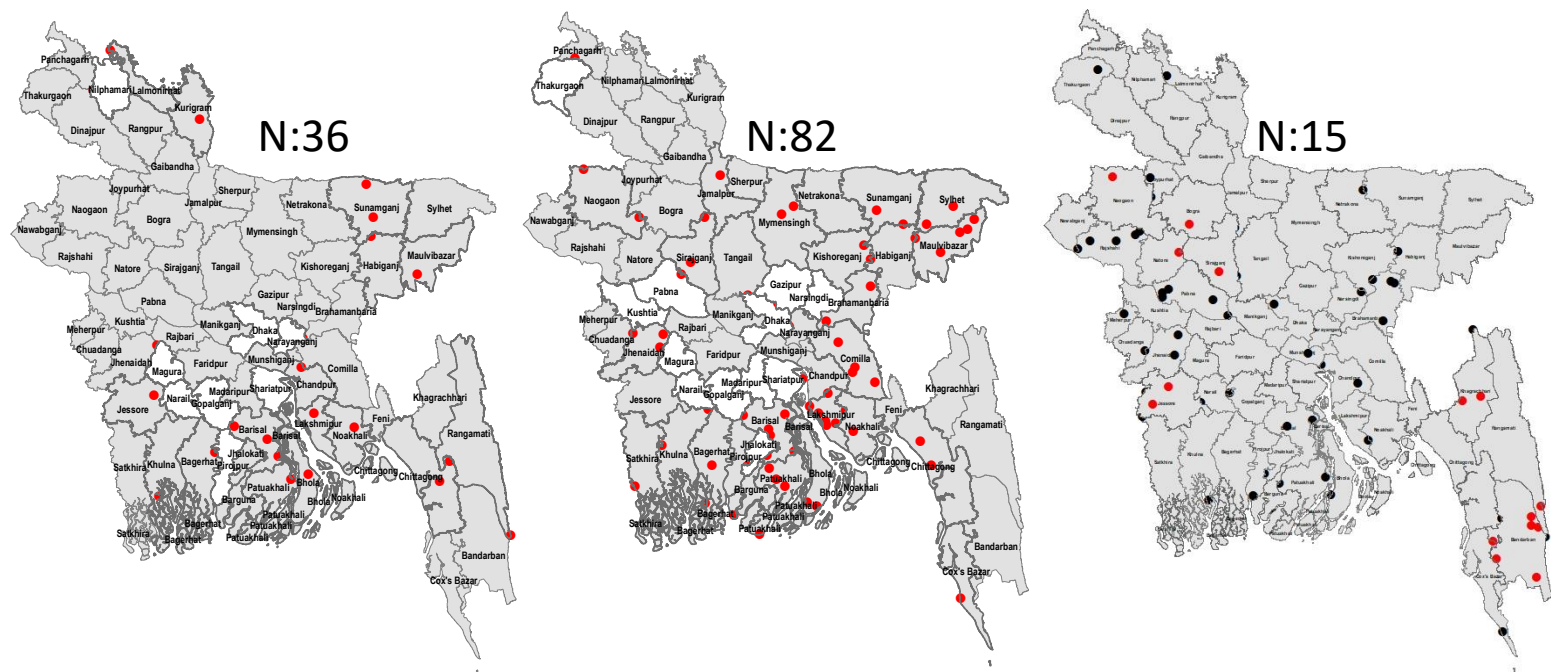
Comparison of Suspected Measles Cases reported in same period (Jan – Sep 2019 & 2020)



Comparison of Suspected Measles Cases reported in same period (Jan – Sep 2019 & 2020)



Distribution of Measles Outbreak 2018-2020*



Confirmed Measles outbreak

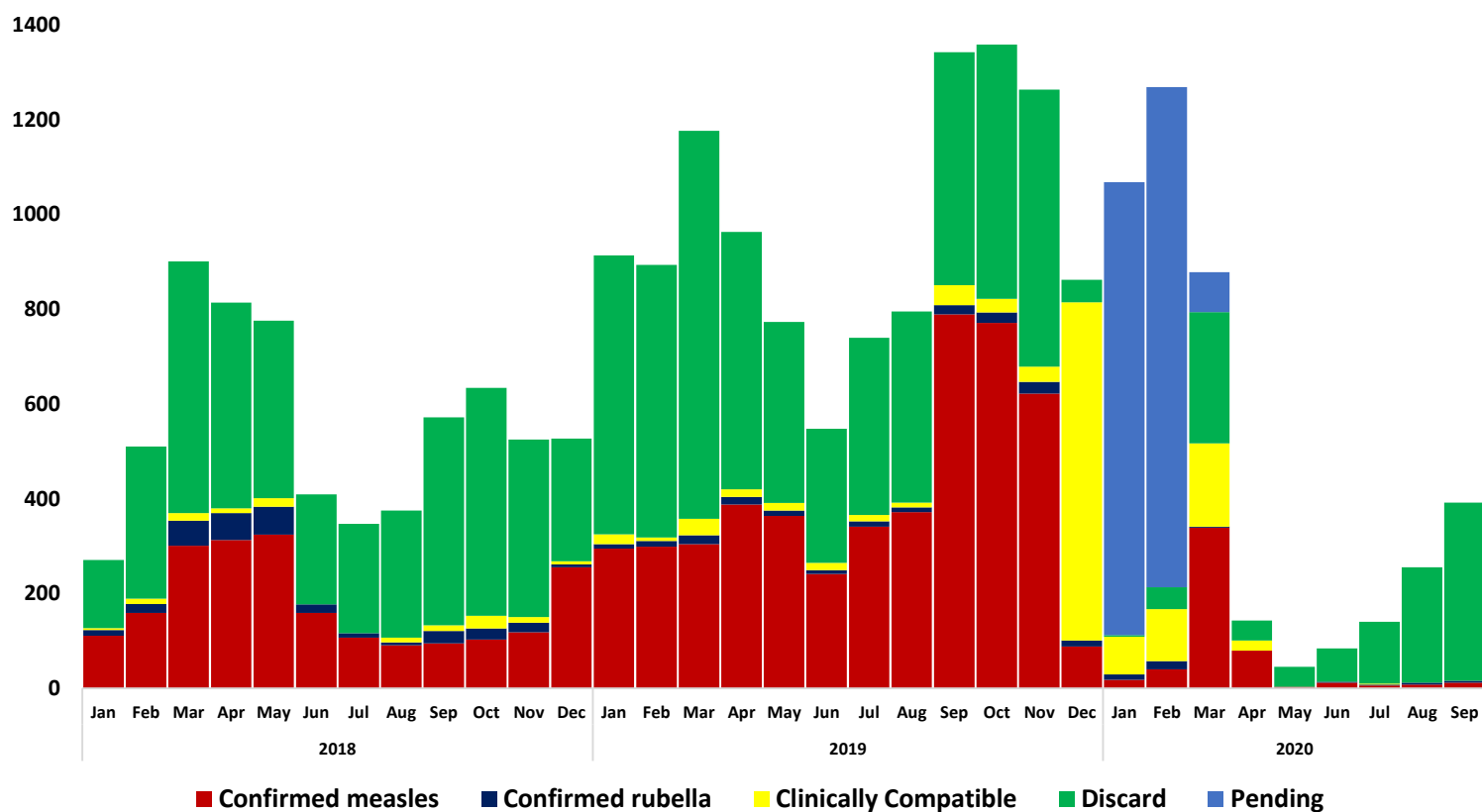
Confirmed Measles outbreak

Confirmed Measles outbreak ●

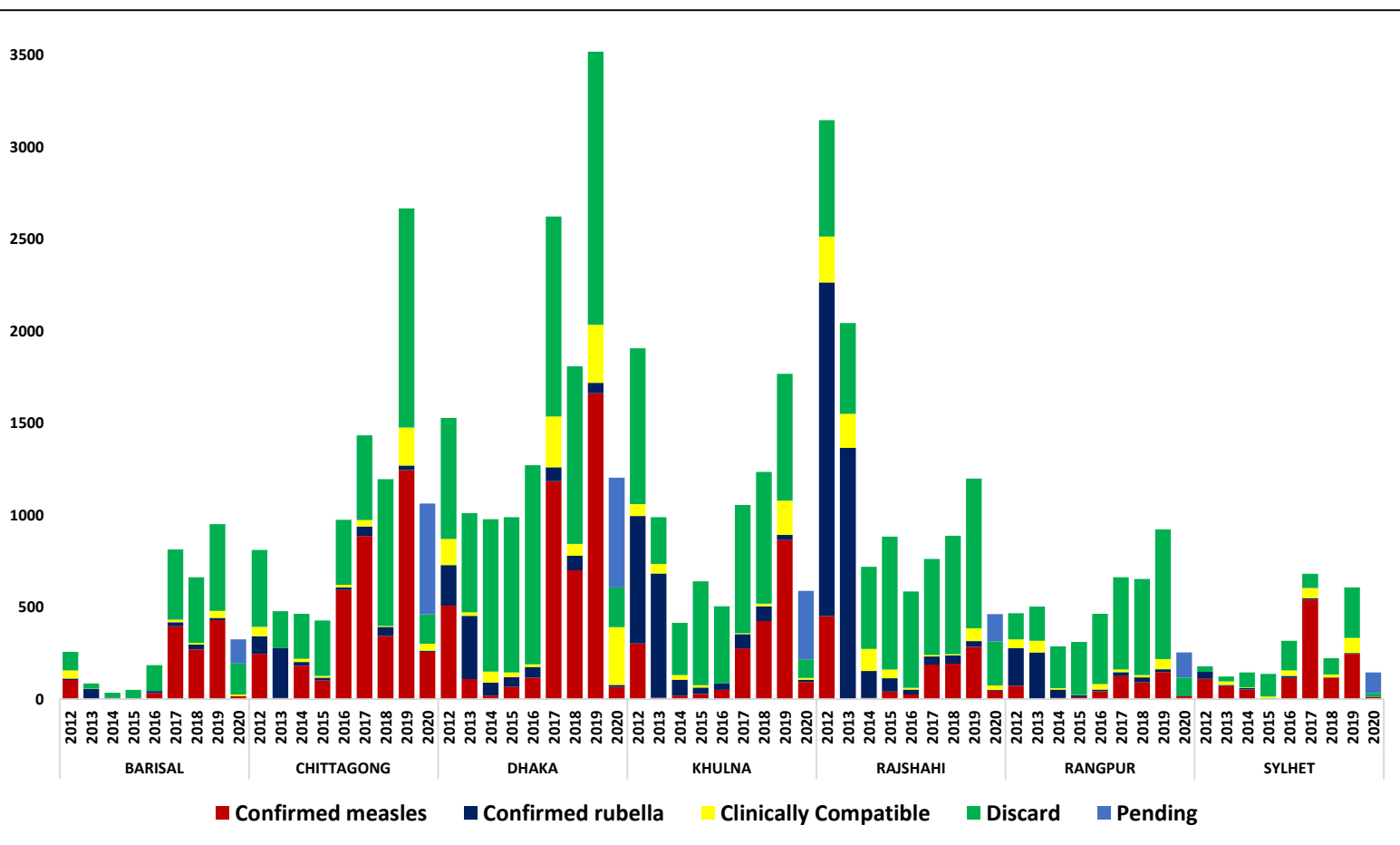
Pending for classification ●

Dots are randomly placed within district

Epicurve by case classification 2018, 2019 & 2020*



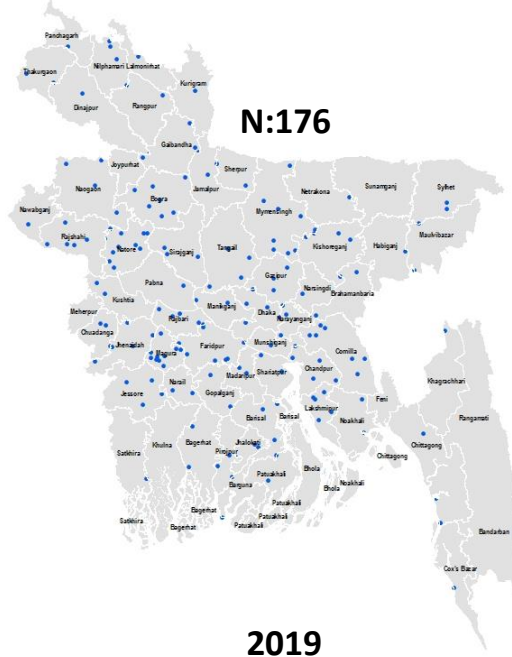
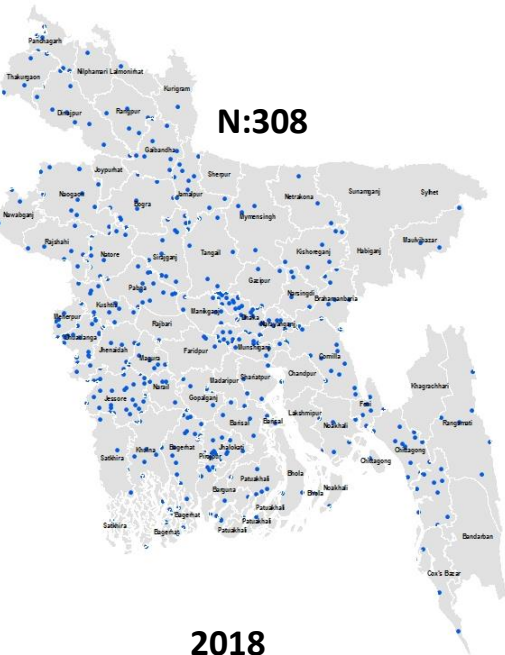
Measles case classification by Division from 2012-2020



Rubella

Distribution of Confirmed Rubella Cases 2018, 2019 & (2020 up to September)

1 dot= 1 case

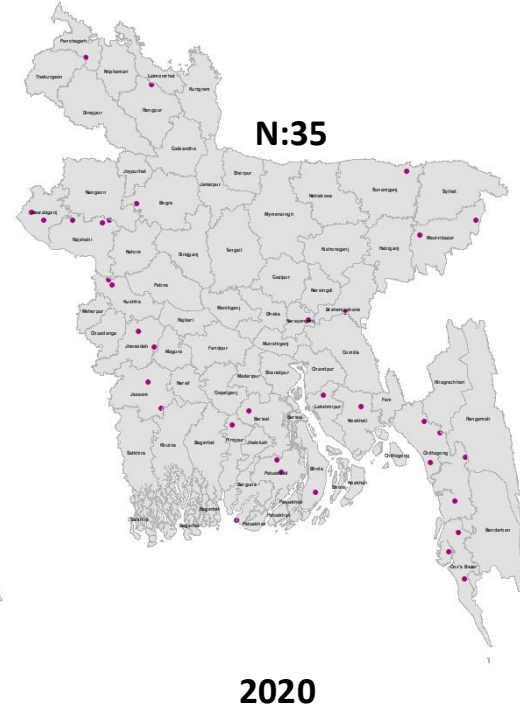
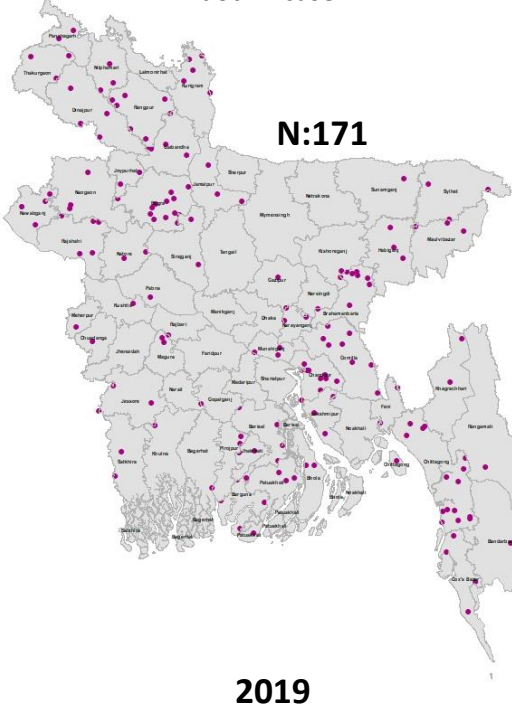
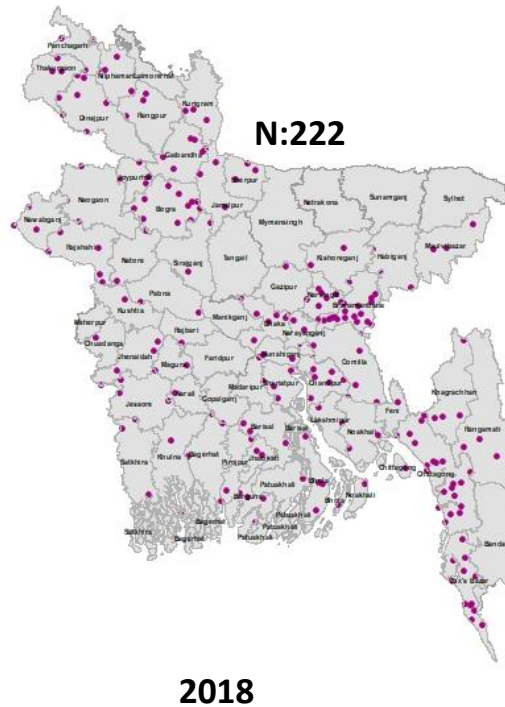


Dots are randomly placed within district

Congenital Rubella Syndrome (CRS)

Distribution of CRS Cases 2018, 2019 & (2020 up to September)

1 dot= 1 case



Dots are randomly placed within district

Summary of outbreaks 2019 & 2020 (Up to September)

Division	District	2019								2020 (Up to September)							
		# Investigated outbreaks	# Measles outbreaks	# Rubella outbreak	# Mixed outbreaks	# Undefined outbreaks	# Discarded outbreaks	# Outbreaks pendings for classification	#Deaths in outbreaks	# Investigated outbreaks	# Measles outbreaks	# Rubella outbreak	# Mixed outbreaks	# Undefined outbreaks	# Discarded outbreaks	# Outbreaks pendings for classification	#Deaths in outbreaks
Barisal	Barguna	1	1	0	0	0	0	0	0	1	0	0	0	0	1	0	0
	Barisal	3	3	0	0	0	0	0	0	1	0	0	0	0	1	0	0
	Barisal CC	3	2	0	0	0	1	0	0	1	0	0	0	0	0	1	0
	Bhola	2	2	0	0	0	0	0	0	1	0	0	0	0	0	1	0
	Jhalakati	3	3	0	0	0	0	0	0	1	0	0	0	0	1	0	0
	Patuakhali	4	4	0	0	0	0	0	0	2	0	0	0	0	2	0	0
	Perojpur	1	0	0	0	0	1	0	0	1	0	0	0	0	1	0	0
	Barisal	17	15	0	0	0	2	0	0	8	0	0	0	0	6	2	0
Chittagong	Bandarban	3	0	0	0	0	2	1	0	6	6	0	0	0	0	0	0
	Brahmanbaria	4	2	0	0	1	1	0	0	4	0	0	0	0	0	4	0
	Chandpur	1	0	0	0	1	0	0	0	2	0	0	0	0	0	2	0
	Chittagong	6	2	0	1	0	3	0	0	0	0	0	0	0	0	0	0
	Chittagong CC	2	0	0	0	1	1	0	0	0	0	0	0	0	0	0	0
	Comilla	8	4	0	0	0	4	0	0	0	0	0	0	0	0	0	0
	Comilla CC	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
	Cox's Bazar	6	1	0	1	1	3	0	0	1	0	0	0	0	0	1	0
	Feni	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
	Khagrachari	0	0	0	0	0	0	0	0	2	2	0	0	0	0	0	1
	Lakshmipur	11	8	0	0	2	0	1	0	1	0	0	0	0	0	1	0
	Noakhali	1	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0
	Rangamati	0	0	0	0	0	0	0	0	1	1	0	0	0	0	0	8
	Chittagong	42	18	0	2	6	14	2	0	17	9	0	0	0	0	8	9
Dhaka	Dhaka	2	0	0	0	0	2	0	3	0	0	0	0	0	0	0	1
	DNCC	5	3	0	0	1	1	0	0	1	0	0	0	1	0	0	0
	DSCC	9	4	0	0	1	4	0	0	0	0	0	0	0	0	0	0
	Faridpur	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
	Gazipur	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
	Gazipur CC	3	1	0	1	0	1	0	0	0	0	0	0	0	0	0	0
	Gopalganj	1	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0
	Kishoreganj	2	2	0	0	0	0	0	0	0	0	0	0	0	0	0	0
	Madaripur	1	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0
	Munshiganj	3	0	0	0	1	2	0	0	1	0	0	0	0	0	1	0
	Narayanganj	1	0	0	0	0	1	0	0	0	0	0	0	0	0	0	0
	Narayanganj CC	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
	Narsingdi	2	1	0	0	1	0	0	0	1	0	0	0	0	0	1	0
	Netrokona	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
	Rajbari	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
	Shariatpur	1	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0
	Tangail	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
	Dhaka	30	14	0	1	4	11	0	3	3	0	0	0	1	0	2	1
Khulna	Bagerhat	2	2	0	0	0	0	0	0	0	0	0	0	0	0	0	0
	Chuadanga	1	0	0	0	1	0	0	0	0	0	0	0	0	0	0	0
	Jessore	1	0	0	0	0	1	0	0	2	2	0	0	0	0	0	0
	Jhenaidah	4	3	0	1	0	0	0	0	3	0	0	0	0	0	3	0
	Khulna	2	2	0	0	0	0	0	0	1	0	0	0	0	0	1	0
	Khulna CC	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
	Kushtia	3	1	0	0	1	1	0	0	0	0	0	0	0	0	0	0
	Magura	6	5	0	0	1	0	0	0	0	0	0	0	0	0	0	0
	Meherpur	2	0	0	0	0	2	0	0	1	0	0	0	0	0	1	0
	Narail	1	1	0	0	0	0	0	0	2	0	0	0	0	0	2	0
	Satkhira	1	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0
	Khulna	23	15	0	1	3	4	0	0	9	2	0	0	0	0	7	0

Summary of outbreaks 2019 & 2020 (Up to September)

Division	District	2019								2020 (Up to September)							
		# Investigated outbreaks	# Measles outbreaks	# Rubella outbreak	# Mixed outbreaks	# Undefined outbreaks	# Discarded outbreaks	# Outbreaks pendings for classification	#Deaths in outbreaks	# Investigated outbreaks	# Measles outbreaks	# Rubella outbreak	# Mixed outbreaks	# Undefined outbreaks	# Discarded outbreaks	# Outbreaks pendings for classification	#Deaths in outbreaks
Mymensingh	Jamalpur	1	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0
	Mymensingh	2	1	0	0	0	1	0	0	0	0	0	0	0	0	0	0
	Mymensingh CC	1	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0
	Netrokona	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
	Sherpur	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
	Mymensingh	4	3	0	0	0	1	0	0	0	0	0	0	0	0	0	0
Rajshahi	Bogra	3	0	0	1	1	1	0	0	1	1	0	0	0	0	0	0
	Joypurhat	1	0	0	0	0	1	0	0	1	0	0	0	0	1	0	0
	Naogaon	12	2	0	0	2	8	0	0	4	1	0	0	0	2	1	0
	Natore	2	0	0	0	1	1	0	0	0	0	0	0	0	0	0	0
	Nowabganj	3	0	0	0	0	3	0	0	0	0	0	0	0	0	0	0
	Pabna	5	1	0	0	2	2	0	0	5	0	0	0	1	4	0	0
	Rajshahi	5	0	0	1	0	4	0	0	1	0	0	0	0	1	0	0
	Rajshahi CC	2	0	0	0	2	0	0	0	4	0	0	0	0	4	0	0
	Sirajganj	11	3	0	1	2	5	0	0	7	2	0	0	0	2	3	0
	Rajshahi	44	6	0	3	10	25	0	0	23	4	0	0	1	14	4	0
Rangpur	Dinajpur	1	0	0	0	0	1	0	0	0	0	0	0	0	0	0	0
	Gaibandha	2	0	0	0	1	1	0	0	0	0	0	0	0	0	0	0
	Kurigram	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
	Lalmonirhat	1	0	0	0	0	1	0	0	1	0	0	0	0	0	1	0
	Nilphamari	6	0	0	1	1	4	0	0	0	0	0	0	0	0	0	0
	Panchagarh	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
	Rangpur	1	0	0	0	0	1	0	0	0	0	0	0	0	0	0	0
	Rangpur CC	1	0	0	1	0	0	0	0	0	0	0	0	0	0	0	0
	Thakurgaon	6	2	0	0	1	3	0	0	1	0	0	0	0	0	1	0
	Rangpur	18	2	0	2	3	11	0	0	2	0	0	0	0	0	2	0
Sylhet	Habiganj	1	0	0	0	0	1	0	0	1	0	0	0	0	1	0	0
	Moulvi Bazar	6	5	0	0	1	0	0	0	0	0	0	0	0	0	0	0
	Sunamganj	4	2	0	0	1	1	0	0	1	0	0	0	0	0	1	0
	Sylhet	3	2	0	0	1	0	0	0	0	0	0	0	0	0	0	0
	Sylhet CC	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
	Sylhet	14	9	0	0	3	2	0	0	2	0	0	0	0	1	1	0
Grand Total		192	82	0	9	29	70	2	3	64	15	0	0	2	21	26	10

Summary of Cases 2019

Division	District	Total cases	Measles Cases						Rubella Cases				Discarded Cases				Cases pending for classification	Incidence		Discarded Rate
			Lab confirmed		Epi-linked	Clinically compatible		Total	Lab confirmed		Epi-linked	Total	Both Negative (Discarded)		Epi-linked (Discarded)	Total		Incidence per 1 million population		Non-measles non-rubella discarded rate per 100,000 pop
			Sporadic cases	Cases from outbreaks	Cases from outbreaks	Sporadic cases	Cases from outbreaks	Total measles cases	Sporadic cases	Cases from outbreaks	Cases from outbreaks	Total rubella cases	Sporadic cases	Cases from outbreaks	Cases from outbreaks	Total discarded cases		Measles	Rubella	District status
BARISAL	BARGUNA	135	59	4	2	4	0	65	1	0	0	1	64	3	0	67	0	60.6	1.0	6.44
	BARISAL	134	35	14	6	5	0	55	2	0	0	2	65	13	0	78	0	21.1	0.9	3.35
	BARISAL CC.	49	9	15	7	1	0	31	1	0	0	1	12	7	4	23	0	62.7	2.6	6.01
	BHOLA	159	37	11	5	7	0	53	0	0	0	0	99	5	0	104	0	23.2	0.0	5.02
	JHALAKATI	143	75	11	2	2	0	88	3	0	0	3	48	4	0	52	0	108.1	3.8	6.54
	PATUAKHALI	249	116	22	9	18	0	147	2	0	0	2	79	9	3	91	0	77.1	1.1	5.09
	PEROJPUR	81	21	0	0	1	0	21	3	0	0	3	48	5	3	56	0	16.2	2.3	4.32
	BARISAL	950	352	77	31	38	0	460	12	0	0	12	415	56	0	471	0	44.2	1.2	4.86
CHITTAGONG	BANDARBAN	103	23	0	0	2	19	23	0	0	0	0	16	9	34	59	0	50.8	0.0	13.04
	BRAHMANBARIA	252	38	45	39	5	15	122	2	0	0	2	83	23	41	147	0	25.1	0.6	4.44
	CHANDPUR	193	93	0	0	14	21	93	3	1	0	4	52	9	0	61	0	33.0	1.4	2.17
	CHITTAGONG	323	100	38	29	17	0	167	2	1	0	3	58	29	78	165	0	23.5	0.5	2.81
	CHITTAGONG CC.	167	89	1	0	15	0	90	0	0	0	0	21	14	27	62	0	29.9	0.0	2.06
	COMILLA	480	106	74	59	14	0	239	6	0	0	6	121	32	127	280	0	30.5	1.0	4.75
	COMILLA CC.	35	13	0	0	1	0	13	1	0	0	1	20	0	0	20	0	34.2	2.6	5.26
	COX'S BAZAR	312	126	42	35	4	8	203	0	1	0	1	52	31	48	131	0	63.0	0.4	4.91
	FENI	102	42	0	0	4	0	42	0	0	0	0	56	0	0	56	0	25.1	0.0	3.34
	KHAGRACHARI	55	17	0	0	1	0	17	1	0	0	1	36	0	0	36	0	23.8	1.4	5.03
	LAKSHMIPUR	343	156	71	40	23	19	267	5	0	0	5	47	22	0	69	0	112.7	2.5	3.42
	NOAKHALI	230	115	7	5	20	0	127	2	0	0	2	83	3	0	86	0	33.1	0.6	2.32
	RANGAMATI	72	48	0	0	5	0	48	0	0	0	0	19	0	0	19	0	69.1	0.0	2.74
	CHITTAGONG	2667	966	278	207	125	82	1451	22	3	0	25	664	527	0	1191	0	37.5	0.8	3.59
DHAKA	DHAKA	286	137	0	0	27	0	137	0	0	0	0	93	14	15	122	0	23.5	0.0	2.09
	DHAKA NORTH CC.	387	239	33	14	29	4	286	1	0	0	1	47	14	20	81	0	59.0	0.2	1.76
	DHAKA SOUTH CC.	465	164	138	120	44	6	422	3	0	0	3	47	36	27	110	0	84.2	0.8	3.07
	FARIDPUR	189	68	0	0	30	0	68	7	0	0	7	84	0	0	84	0	30.5	3.1	3.77
	GAZIPUR	97	44	0	0	5	0	44	0	0	0	0	48	0	0	48	0	20.8	0.0	2.27
	GAZIPUR CC.	125	38	10	5	26	0	53	3	1	0	4	27	9	11	47	0	25.9	2.2	2.53
	GOPALGANJ	178	82	29	27	16	0	138	2	0	0	2	47	2	0	49	0	81.3	1.5	3.59
	KISHOREGANJ	160	48	16	5	21	0	69	4	0	0	4	69	2	0	71	0	18.9	1.2	2.09
	MADARIPUR	131	58	9	7	11	0	74	2	0	0	2	48	3	0	51	0	49.3	1.5	3.75
	MANIKGANJ	78	20	0	0	6	0	20	2	0	0	2	50	0	0	50	0	12.3	1.2	3.08
	MUNSHIGANJ	140	36	1	0	3	0	37	2	0	0	2	23	19	56	98	0	22.0	1.2	5.82
	NARAYANGANJ	169	74	0	0	9	0	74	3	0	0	3	80	3	0	83	0	28.4	1.2	3.18
	NARAYANGANJ CC.	14	11	0	0	0	0	11	0	0	0	0	3	0	0	3	0	13.3	0.0	0.36
	NARSINGDI	128	44	5	0	2	0	49	1	0	0	1	67	9	0	76	0	18.9	0.4	2.93
	RAJBARI	106	29	0	0	8	0	29	5	0	0	5	64	0	0	64	0	23.7	4.1	5.23
	SARIATPUR	102	18	23	17	7	0	58	2	0	0	2	52	0	0	52	0	30.4	1.5	3.86
	TANGAIL	199	92	0	0	13	0	92	5	0	0	5	89	0	0	89	0	21.9	1.2	2.12
	DHAKA	2954	1202	264	195	257	10	1661	42	1	0	43	938	240	0	1178	0	34.5	1.0	2.77
KHULNA	BAGERHAT	193	78	26	18	19	0	122	2	0	0	2	66	2	0	68	0	61.0	1.2	3.90
	CHUADANGA	120	51	1	0	3	12	52	2	0	0	2	47	4	0	51	0	39.5	1.5	3.88
	JESSORE	140	57	0	0	18	0	57	4	0	0	4	54	5	2	61	0	17.7	1.2	1.89
	JHENAIDAH	260	94	77	67	15	1	238	5	1	0	6	58	9	0	67	0	82.9	2.9	3.25
	KHULNA	133	61	10	0	12	0	71	0	0	0	0	47	3	0	50	0	36.8	0.0	2.59
	KHULNA CC.	31	20	0	0	4	0	20	0	0	0	0	7	0	0	7	0	25.9	0.0	0.91
	KUSHTIA	210	65	35	30	9	3	130	1	1	0	2	76	9	11	96	0	44.1	0.9	4.23
	MAGURA	367	88	122	108	26	23	318	9	1	0	10	83	15	0	98	0	196.2	9.3	9.16
	MEHERPUR	78	18	0	0	0	0	18	0	0	0	0	29	10	21	60	0	23.6	0.0	7.86
	NARAIL	143	26	9	7	28	0	42	2	0	0	2	75	3	0	78	0	41.6	2.4	9.28
	SATKHIRA	93	20	6	2	13	0	28	1	0	0	1	51	2	0	53	0	11.2	0.4	2.29
	KHULNA	1768	578	286	232	147	39	1096	26	3	0	29	593	96	0	689	0	47.3	1.6	3.76

Summary of Cases 2019

Division	District	Total cases	Measles Cases						Rubella Cases				Discarded Cases				Cases pending for classification	Incidence		Discarded Rate
			Lab confirmed		Epi-linked	Clinically compatible		Total	Lab confirmed		Epi-linked	Total	Both Negative (Discarded)		Epi-linked (Discarded)	Total		Incidence per 1 million population		Non-measles non-rubella discarded rate per 100,000 pop
			Sporadic cases	Cases from outbreaks	Cases from outbreaks	Sporadic cases	Cases from outbreaks	Total measles cases	Sporadic cases	Cases from outbreaks	Cases from outbreaks	Total rubella cases	Sporadic cases	Cases from outbreaks	Cases from outbreaks	Total discarded cases		Measles	Rubella	District status
MYMENSINGH	JAMALPUR	115	27	6	0	3	0	33	1	0	0	1	78	0	0	78	0	12.4	0.4	2.92
	MYMENSINGH	258	84	8	5	33	0	97	9	0	0	9	116	8	0	124	0	16.3	1.6	2.19
	MYMENSINGH CC.	50	21	20	16	5	0	57	1	0	0	1	3	0	0	3	0	136.4	3.3	1.00
	NETROKONA	89	20	0	0	6	0	20	1	0	0	1	62	0	0	62	0	7.7	0.4	2.39
	SHERPUR	53	10	0	0	1	0	10	2	0	0	2	40	0	0	40	0	6.3	1.3	2.53
	MYMENSINGH	565	162	34	21	48	0	217	14	0	0	14	299	8	0	307	0	15.3	1.1	2.40
RAJSHAHI	BOGRA	143	18	2	0	2	0	20	6	1	0	7	85	29	0	114	0	5.0	1.8	2.88
	JOYPURHAT	47	6	0	0	4	0	6	0	0	0	0	33	4	0	37	0	5.6	0.0	3.48
	NATORE	149	28	1	0	3	7	29	10	0	0	10	79	12	9	100	0	14.6	5.0	5.03
	NOAGOAN	120	10	4	0	4	4	14	2	2	0	4	48	46	0	94	0	4.6	1.3	3.10
	NOWABGANJ	85	14	0	0	5	0	14	1	0	0	1	39	21	5	65	0	7.3	0.5	3.39
	PABNA	240	57	8	3	8	22	68	1	0	0	1	88	21	35	144	0	22.1	0.3	4.90
	RAJSHAHI	131	7	6	4	4	0	17	4	1	0	5	52	23	34	109	0	5.2	2.0	4.36
	RAJSHAHI CC.	37	1	1	0	2	4	2	0	1	0	1	20	8	0	28	0	3.8	1.9	5.34
	SIRAJGANJ	247	26	94	78	1	0	198	1	2	0	3	51	41	31	123	0	33.2	0.8	3.41
	RAJSHAHI	1199	167	116	85	33	37	368	25	7	0	32	495	319	0	814	0	13.1	1.5	3.78
RANGPUR	DINAJPUR	158	17	0	0	12	0	17	2	0	0	2	111	5	11	127	0	4.9	0.6	3.65
	GAIBANDHA	130	11	1	0	7	8	12	4	0	0	4	69	19	11	99	0	4.3	1.4	3.57
	KURIGRAM	115	20	0	0	5	0	20	1	0	0	1	89	0	0	89	0	8.3	0.4	3.69
	LALMONIRHAT	50	10	0	0	2	0	10	1	0	0	1	32	5	0	37	0	6.8	0.7	2.53
	NILPHAMARI	116	7	1	0	3	2	8	0	3	0	3	60	25	15	100	0	3.7	1.4	4.68
	PANCHAGHARH	49	9	0	0	2	0	9	1	0	0	1	37	0	0	37	0	7.8	0.9	3.22
	RANGPUR	89	14	0	0	7	0	14	0	0	0	0	56	3	9	68	0	4.6	0.0	2.26
	RANGPUR CC.	27	7	1	0	0	0	8	1	1	0	2	8	9	0	17	0	23.3	5.8	4.96
	THAKURGOAN	188	34	14	8	1	6	56	2	0	0	2	66	35	30	131	0	29.6	1.2	8.09
	RANGPUR	922	129	17	8	39	16	154	12	4	0	16	528	177	0	705	0	7.9	0.9	3.83
SYLHET	HABIGANJ	102	25	0	0	24	0	25	1	0	0	1	34	4	14	52	0	10.3	0.4	2.14
	MOULVI BAZAR	156	46	22	6	13	21	74	1	0	0	1	47	6	0	53	0	30.4	0.4	2.37
	SUNAMGANJ	188	62	27	16	14	3	105	0	0	0	0	59	12	11	82	0	30.9	0.0	2.85
	SYLHET	142	54	6	0	4	0	60	2	0	0	2	54	7	15	76	0	18.0	0.6	2.27
	SYLHET CC.	19	4	0	0	2	0	4	1	0	0	1	12	0	0	12	0	7.2	1.8	2.15
	SYLHET	607	191	55	22	57	24	268	5	0	0	5	206	69	0	275	0	21.5	0.4	2.40
	Grand Total	11632	3747	1127	801	744	208	5675	158	18	0	176	4138	1492	0	5630	0	29.0	1.05	3.35

Summary of Cases 2020

(For 2020 all rates calculated up to Epidemiological week 39)

Division	District	Total cases	Measles Cases			Clinically compatible	Rubella Cases			Discarded Cases	Cases pending for classification	Incidence		Discarded Rate
			Lab confirmed	Epi-linked	Total		Lab confirmed	Epi-linked	Total			Both Negative (Discarded)	Incidence per 1 million population	
										Measles			Rubella	District status
BARISAL	BARGUNA	54	2	0	2	1	0	0	0	38	13	2.40	0.00	4.57
	BARISAL	54	1	0	1	2	0	0	0	35	16	0.54	0.00	1.88
	BARISAL CC.	19	0	0	0	1	0	0	0	8	10	0.00	0.00	2.61
	BHOLA	64	4	0	4	1	0	0	0	29	30	2.41	0.00	1.75
	JHALAKATI	25	2	0	2	1	0	0	0	12	10	3.14	0.00	1.89
	PATUAKHALI	95	6	0	6	1	0	0	0	46	42	4.19	0.00	3.21
	PEROJPUR	41	1	0	1	1	1	0	1	28	10	0.96	0.96	2.70
	BARISAL	352	16	0	16	8	1	0	1	196	131	2.06	0.13	2.53
CHITTAGONG	BANDARBAN	97	25	39	64	2	0	0	0	15	16	176.79	0.00	4.14
	BRAHMANBARIA	191	4	0	4	2	1	0	1	16	168	1.51	0.38	0.60
	CHANDPUR	101	8	0	8	11	0	0	0	26	56	3.55	0.00	1.15
	CHITTAGONG	65	15	0	15	0	1	0	1	13	36	3.20	0.21	0.28
	CHITTAGONG CC.	78	20	0	20	0	0	0	0	16	42	8.30	0.00	0.66
	COMILLA	81	9	0	9	8	1	0	1	34	29	1.91	0.21	0.72
	COMILLA CC.	5	0	0	0	0	0	0	0	2	3	0.00	0.00	0.66
	COX'S BAZAR	128	7	0	7	1	1	0	1	12	107	3.28	0.47	0.56
	FENI	18	0	0	0	0	0	0	0	8	10	0.00	0.00	0.60
	KHAGRACHARI	132	11	92	103	0	0	0	0	24	5	179.97	0.00	4.19
	LAKSHMIPUR	102	19	0	19	6	0	0	0	4	73	11.79	0.00	0.25
	NOAKHALI	66	6	0	6	7	2	0	2	7	44	2.07	0.69	0.24
	RANGAMATI	25	4	0	4	0	0	0	0	9	12	7.20	0.00	1.62
	CHITTAGONG	1089	128	131	259	37	6	0	6	186	601	9.77	0.23	0.70
DHAKA	DHAKA	86	4	0	4	32	1	0	1	9	40	0.86	0.21	0.19
	DHAKA NORTH CC.	197	5	0	5	86	1	0	1	37	68	1.36	0.27	1.00
	DHAKA SOUTH CC.	181	8	0	8	81	2	0	2	15	75	2.79	0.70	0.52
	FARIDPUR	79	5	0	5	3	1	0	1	4	66	2.80	0.56	0.22
	GAZIPUR	57	5	0	5	12	2	0	2	14	24	2.96	1.18	0.83
	GAZIPUR CC.	40	4	0	4	11	3	0	3	3	19	2.70	2.02	0.20
	GOPALGANJ	47	3	0	3	4	0	0	0	18	22	2.74	0.00	1.65
	KISHOREGANJ	41	1	0	1	7	0	0	0	9	24	0.37	0.00	0.33
	MADARIPUR	46	3	0	3	7	1	0	1	19	16	2.76	0.92	1.75
	MANIKGANJ	19	0	0	0	2	0	0	0	6	11	0.00	0.00	0.46
	MUNSHIGANJ	48	1	0	1	14	0	0	0	4	29	0.74	0.00	0.30
	NARAYANGANJ	61	2	0	2	27	0	0	0	17	15	0.96	0.00	0.81
	NARAYANGANJ CC.	4	0	0	0	0	0	0	0	1	3	0.00	0.00	0.15
	NARSINGDHI	66	7	0	7	11	0	0	0	20	28	3.37	0.00	0.96
	RAJBARI	32	2	0	2	1	1	0	1	11	17	2.04	1.02	1.12
	SARIATPUR	41	2	0	2	3	0	0	0	26	10	1.86	0.00	2.41
	TANGAIL	72	7	0	7	4	2	0	2	19	40	2.08	0.60	0.57
DHAKA	1117	59	0	59	305	14	0	14	232	507	1.74	0.41	0.68	
KHULNA	BAGERHAT	60	4	0	4	0	0	0	0	22	34	2.91	0.00	1.60
	CHUADANGA	13	2	0	2	0	1	0	1	2	8	1.90	0.95	0.19
	JESSORE	116	26	16	42	3	4	0	4	24	43	16.30	1.55	0.93
	JHENAIDAH	64	15	2	17	1	0	0	0	8	38	10.30	0.00	0.48
	KHULNA	30	3	0	3	0	0	0	0	12	15	1.94	0.00	0.78
	KHULNA CC.	6	0	0	0	0	0	0	0	0	6	0.00	0.00	0.00
	KUSHTIA	65	7	0	7	2	0	0	0	17	39	3.86	0.00	0.94
	MAGURA	46	4	0	4	2	0	0	0	13	27	4.67	0.00	1.52
	MEHERPUR	43	0	0	0	1	0	0	0	5	37	0.00	0.00	0.82
	NARAIL	125	11	0	11	0	4	0	4	12	98	16.35	5.95	1.78
	SATKHIRA	40	6	0	6	0	3	0	3	6	25	3.24	1.62	0.32
KHULNA	608	78	18	96	9	12	0	12	121	370	6.56	0.82	0.83	

Summary of Cases 2020

(For 2020 all rates calculated up to Epidemiological week 39)

Division	District	Total cases	Measles Cases			Clinically compatible	Rubella Cases			Discarded Cases	Cases pending for classification	Incidence		Discarded Rate
			Lab confirmed	Epi-linked	Total		Lab confirmed	Epi-linked	Total			Both Negative (Discarded)	Incidence per 1 million population	
										Measles			Rubella	District status
MYMENSINGH	JAMALPUR	29	1	0	1	1	0	0	0	8	19	0.47	0.00	0.37
	MYMENSINGH	66	4	0	4	7	0	0	0	8	47	0.88	0.00	0.18
	MYMENSINGH CC.	4	0	0	0	0	0	0	0	0	4	0.00	0.00	0.00
	NETROKONA	34	4	0	4	0	0	0	0	18	12	1.92	0.00	0.87
	SHERPUR	10	0	0	0	0	0	0	0	4	6	0.00	0.00	0.32
	MYMENSINGH	143	9	0	9	8	0	0	0	38	88	0.88	0.00	0.37
RAJSHAHI	BOGRA	45	4	0	4	0	0	0	0	25	16	1.26	0.00	0.79
	JOYPURHAT	22	1	0	1	0	0	0	0	11	10	1.17	0.00	1.29
	NATORE	48	6	0	6	0	1	0	1	29	12	3.77	0.63	1.82
	NOAGOAN	47	3	2	5	1	1	0	1	22	18	2.06	0.41	0.91
	NOWABGANJ	30	4	0	4	1	1	0	1	15	9	2.60	0.65	0.98
	PABNA	127	4	0	4	20	0	0	0	88	15	1.70	0.00	3.74
	RAJSHAHI	33	2	0	2	0	1	0	1	18	12	1.00	0.50	0.90
	RAJSHAHI CC.	46	2	0	2	0	0	0	0	37	7	4.77	0.00	8.82
	SIRAJGANJ	107	9	12	21	1	0	0	0	35	50	7.27	0.00	1.21
	RAJSHAHI	505	35	14	49	23	4	0	4	280	149	2.84	0.23	1.62
RANGPUR	DINAJPUR	44	0	0	0	0	2	0	2	20	22	0.00	0.72	0.72
	GAIBANDHA	26	2	0	2	0	0	0	0	11	13	0.90	0.00	0.50
	KURIGRAM	54	4	0	4	0	0	0	0	30	20	2.07	0.00	1.56
	LALMONIRHAT	11	0	0	0	0	0	0	0	1	10	0.00	0.00	0.09
	NILPHAMARI	42	2	0	2	0	0	0	0	18	22	1.17	0.00	1.05
	PANCHAGHARH	20	1	0	1	0	0	0	0	10	9	1.09	0.00	1.09
	RANGPUR	25	5	0	5	0	0	0	0	11	9	2.07	0.00	0.46
	RANGPUR CC.	3	1	0	1	0	0	0	0	0	2	3.65	0.00	0.00
	THAKURGOAN	65	1	0	1	0	2	0	2	31	31	0.77	1.54	2.39
	RANGPUR	290	16	0	16	0	4	0	4	132	138	1.09	0.27	0.90
SYLHET	HABIGANJ	39	1	0	1	0	0	0	0	35	3	0.51	0.00	1.80
	MOULVI BAZAR	35	1	0	1	0	1	0	1	2	31	0.56	0.56	0.11
	SUNAMGANJ	72	10	0	10	0	0	0	0	5	57	4.35	0.00	0.22
	SYLHET	25	1	0	1	0	0	0	0	5	19	0.36	0.00	0.18
	SYLHET CC.	2	0	0	0	0	0	0	0	0	2	0.00	0.00	0.00
	SYLHET	173	13	0	13	0	1	0	1	47	112	1.41	0.11	0.51
	Grand Total	4277	354	163	517	390	42	0	42	1232	2096	3.85	0.31	0.92

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Key Definitions

Key definitions

- **Suspected measles and rubella:** Any person in whom a clinician suspects measles infection or any person with fever and generalized maculopapular (i.e. non-vesicular) rash and any of the following:
 - Cough
 - Coryza (runny nose)
 - Conjunctivitis (red eyes)
- **Suspected CRS:** Any infant less than one year of age in whom a health worker suspects CRS:
 - A health worker should suspect CRS when an infant's mother has a history of suspected or confirmed rubella during pregnancy, even when the infant shows no signs of CRS.
 - A health worker should also suspect CRS when an infant aged 0-11 month presents with heart disease and/or suspicion of deafness and/or one or more of the following eye signs: white pupil (cataract); diminished vision; pendular movement of the eyes (nystagmus); squint; small eye ball (microphthalmos) or larger eye ball (congenital glaucoma)
- **Laboratory-confirmed measles, rubella and CRS case:** A case that meets the case definition and has laboratory confirmation of measles/rubella infection and/or isolation of virus in a clinical sample by WHO accredited national laboratory.
- **Epidemiologically-linked confirmed measles, or rubella, case:** A case that meets the clinical case definition and is linked epidemiologically to a laboratory confirmed case or another epidemiologically confirmed case.
- **Clinically compatible measles:** Any person meeting the clinical case definition in whom a laboratory investigation is not conducted and does not have epidemiologic linkage to a laboratory confirmed case.
- **Clinically compatible rubella:** Not applicable.
- Infant in whom a qualified physician detects at least two of the complications listed in (A) or one in group (A) and one in group (B).
- Cataract and or congenital glaucoma; congenital heart disease; loss hearing; pigmentary retinopathy.
- Purpura, splenomegaly; microcephaly; mental retardation; meningoencephalitis; radiolucent bone disease; jaundice with onset within 24 hours after birth.
- **Measles outbreak:** an occurrence of 3 or more suspected measles cases in one month in a rural ward/urban mahalla.
- **Rubella outbreak:** N/A
- **Non-measles non-rubella discarded case:** A suspect case that has been subjected to serology or virus identification but has not met any of the criteria for diagnosis, or a suspected case which has been linked epidemiologically to a confirmed case of another communicable disease.
- **Measles Incidence:** Annualized Incidence of Confirmed Measles (Laboratory and Epidemiologically confirmed) cases per million population.
Calculation: $\# \text{ of confirmed measles case} \div \text{Total population} * 10,00,000 * (52/\text{Epidemiological Week})$
- **Discarded Rate :** Annualized Reporting rate of non-measles non-rubella cases at national level
Calculation: $\# \text{ of discarded case} \div \text{Total population} * 1,00,000 * (52/\text{Epidemiological Week})$

